

Verification Continuum™

Verdi® Supplementary User Guide

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Introduction

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What is Verdi Supplementary Package?

Verdi Supplementary Package is a special package for the users who do not have VCS installed. You can use **Vericom/Vhdlcom Advanced Mode** after installing it inside Verdi installation.

What is Vericom/Vhdlcom Advanced Mode?

Vericom/Vhdlcom Advanced Mode is for vericom/vhdlcom users to transfer parser to vlogan with one modification in the compiling script. In **Vericom/Vhdlcom Advanced Mode**, you can have privilege of getting support of features up-to-date, and it also provides better consistency on behaviors between KDB and simulator. You can have better performance by using **Vericom/Vhdlcom Advanced Mode**, one benefit is shorter parsing time, the other important advantage is that you are able to use **Verdi Smart Load** and **PreLoad** capabilities, which reduce lots of memory and loading time especially for a huge design.

Release Requirement - Prerequisites

It is recommended to use *Verdi* and *VCS* (if you can access *VCS* in your environment) with versions newer than below:

VCS: T-2022.06-SP2-1 ~ U-2023.03-SP1-1, Linux 64-Bit.

VCS: U-2023.03-SP2 or later version, Linux 32/64-Bit, AArch64.

Verdi: T-2022.06-SP2-1 ~ U-2023.03-SP1-1, Linux 64-Bit.

Verdi: U-2023.03-SP2 or later version, Linux 32/64-Bit, AArch64.

Who Should Download Verdi Supplementary Package?

You will need to download and install the extra **Verdi Supplementary Package** in below two conditions:

- You are using third party simulators and has no VCS simulator installed (or has no accessibility to VCS).
- You can access VCS; however, the VCS version is older than T-2022.06-SP2-1, and it cannot be updated.

Check the *Verdi Supplementary Installation Guide* to get and install the package, if your environment is the same as above.

If you have installed VCS or have the access to VCS in your environment, you can directly use that VCS to execute the **Vericom/Vhdlcom Advance Mode**, there is no need to download any extra package in this case.

How to use This Document – User Map

Find the place you are and where you need to go.

- Third party simulator users.
 - Using vericom as KDB compiler.
 - Check the following sessions:
 - [Using Vericom/Vhdlcom Advanced Mode](#)
 - [Fault Tolerance in Vericom \(vhdlcom does not support\)](#)
 - [Elaboration and Smart Load](#)
 - Using UFE flow as KDB compiler.

- You can continue to use UFE as the compiler; however, there may exist some inconsistent between your KDB and FSDB.
- If you want to switch to *Vericom* flow to avoid inconsistency, check the following sessions:
 - [Using Vericom/Vhdlcom Advanced Mode](#)
 - [Fault Tolerance in Vericom \(vhdlcom does not support\)](#)
 - [Elaboration and Smart Load](#)
- VCS simulator users.
 - Using vericom as KDB compiler.
 - It is recommended to use UFE (vlogan or vcs) flow to replace your existing Vericom flow. However, if you wish to continue using Vericom, check the following sessions:
 - [Using Vericom/Vhdlcom Advanced Mode](#)
 - [Fault Tolerance in Vericom \(vhdlcom does not support\)](#)
 - [Elaboration and Smart Load](#)
 - Using UFE flow as KDB compiler.
 - Continue to use UFE.

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Overview

If VCS is installed

You can use Vericom/Vhdlcom Advanced Mode without by simply setting up `VCS_HOME` and point to the VCS installed directory.

```
% setenv VCS_HOME <VCS_installed_directory>
```

If VCS is not installed

You need to install Verdi Supplementary package under path such as:

```
$VERDI_HOME/platform/
```

In this chapter:

- [Using Vericom/Vhdlcom Advanced Mode](#)
- [Passing Extra Related Options to Advanced Mode](#)
- [Fault Tolerance in Vericom \(vhdlcom does not support\)](#)
- [Elaboration and Smart Load](#)

Using Vericom/Vhdlcom Advanced Mode

After setting up `VCS_HOME` or installing the Verdi supplementary package, you can use it by applying switches in the command line:

Command-Line Switches

vericom/vhdlcom	-usevcs=adv	VCS flow
	-usemti=adv	MTI flow
	-useius=adv	IUS flow
Option <code>-kdb=common_elab</code> and <code>-full164</code> will be added in vlogan automatically.		

Passing Extra Related Options to Advanced Mode

Command-Line Switches

vericom/vhdlcom/elabcom	-usevcs=adv -XOptions="... "	VCS flow
	-usemti=adv -XOptions="..."	MTI flow
	-useius=adv -XOptions="..."	IUS flow

Example:

```
vericom -usevcs=adv -XOptions="-timescale=1ns/1ns"  
elabcom -usevcs=adv -XOptions="-wreal res_def+scalarize+coerce"
```

Fault Tolerance in Vericom (vhdlcom does not support)

Note:

Fault tolerance is not supported in Vhdlcom.

Fault tolerance is doing the best to parse the error design and generate the database, which gives the flexibility on errors like vericom.

Usage

```
vericom -usevcs=adv -ft
```

Example: test.v

```
module test;
int x y; // Should be separated by comma ","
endmodule
%> vericom -usevcs=adv -ft test.v
.....
Parsing design file 'test.v'
Error-[SE] Syntax error
  Following verilog source has syntax error :
    Illegal port or variable declaration
    "test.v", 3: token is 'y'
    int x y;
        ^
Error-[UTOPN] Unknown type or port name
  The type name 'int' is unknown, or the identifier 'x' has not been
  listed as
  a port, or the declaration might represent an instance missing
  parentheses.
  "test.v", 3
  Source info: int x y;
3 errors | Errors found at first pass
Re-Parsing Design with Fault Tolerance
.....
Parsing design file 'test.v'
CPU time: .366 seconds to compile | Passed after re-compile
```

Limitation and Known Issues

The syntax errors that possibly make the scope cannot be created successfully.

1. Matching semicolon ';' across module scopes, it will make those module scopes not in skip list, once they also have vital syntax errors, they will not be generated.
2. Matching brace) for (across module scopes, just like previous, once those module scopes not in skip list also have vital syntax errors, they will not be generated.

Fault Tolerance in Vericom (vhdlcom does not support)

3. Matching 'endmodule' for 'module' across module scopes, just like previous, once those module scopes not in skip list also have vital syntax errors, they will not be generated.
4. Errors that cannot be suppressed by vlogan now, here are two examples:
 - a. IPD (Identifier previously declared)
 - b. IND (Identifier not declared)

Elaboration and Smart Load

You can use Smart Load after elaboration, smart load can save massive memory and open design much more quickly especially for a huge design.

Before 2023.03

```
> setenv NOVAS_EXECUTABLE_ARCH 64
> elabcom -usevcs=adv -lib work
> verdi -usevcs=adv -smart_load_kdb -elab kdb.elab++
```

After 2023.03

```
> setenv NOVAS_EXECUTABLE_ARCH 64
> elabcom -usevcs=adv -lib work -top system
> verdi -usevcs=adv -smart_load_kdb -elab kdb.elab++
or
> verdi -usevcs=adv -smart_load_kdb -dbdir simv.daidir
```

After 2023.03-SP2

```
> setenv NOVAS_EXECUTABLE_ARCH 32/64
> elabcom -usevcs=adv -lib work -top system
> verdi -usevcs=adv -smart_load_kdb -elab kdb.elab++
or
> verdi -usevcs=adv -smart_load_kdb -dbdir simv.daidir
```

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Appendix

In this chapter:

- [Vericom to Vlogan Options Mapping Table](#)
- [Vhdlcom to Vhdlan Options Mapping Table](#)
- [Elabcom to VCS Elaboration Options Mapping Table](#)
- [Libraries Mapping](#)

Vericom to Vlogan Options Mapping Table

All vericom options mapped for vlogan would be allowed as following:

VERICOM OPTIONS	DESCRIPTION	VLOGAN OPTIONS
-XOptions="..."	For vericom/vhdlcom/elabcom in advanced mode flow to pass the related options without mapped.	Related vlogan/vhdlan/vcs options
-2001	Use Verilog IEEE 1364-2001 naming style for generate	+v2k
-2001genblk	blocks (overrides other language options).	+v2k
-2005	Support Verilog IEEE 1364-2005 standard.	-v2005
-2009	Support Verilog IEEE 1800-2009 standard.	-sverilog
-2012	Support Verilog IEEE 1800-2012 standard.	-sverilog
-ams	Support Verilog-AMS syntax.	-ams +incdir+ \$ENV{BT_ROOT}/kdb/etc/v erilog/ams
-assert checker svaext svvunit	Specify the additional syntax to support. checker: Support the checker construct of SystemVerilog IEEE 1800-2009 standard. svaext: Support SystemVerilog Assertion features compliant to IEEE 1800-2009 standard. svvunit: Support PSL vunit syntax. Files with the *.psl file extension will be treated as PSL files.	-assert checker svaext svvunit
-applog	Append the compile messages to log file (The default mode is overwriting).	

<code>-autoendcelldef</code>	Automatically appends <code>`endcelldefine</code> at the end of a file if a matching <code>`endcelldefine</code> cannot be found for a declared <code>`celldefine</code> in the file.	
<code>-chmod <numeric_mode></code>	Specify the access attribute for the generated files and directories.	
<code>-comment_transoff_regions</code> <code>-suboption +suboption</code>	Skip source code between <code>translate_off/on</code> pragmas The suboptions are vendor names e.g., cadence, ikos, mentor, novas, pragma, quickturn, synopsys, synthesis.	<code>-skip_translate_body</code> <code>=pragma+...</code>
<code>-cuname <compilation-unit name></code>	Support compilation-based compilation-unit mode with specific name. The global space will be modeled as a package named "compilation-unit name". It will be disabled if specified with <code>-cunit</code> .	
<code>-cunit</code>	Support compilation-based compilation-unit mode with default name. The global space will be modeled as a package named "novas_cunit_n"	
<code>+define+<macro></code>	The <code>+define</code> option is used to specify macros. If the macro is also defined in your source code, it will be overridden by this option.	<code>+define+<marco></code>
<code>-define <macro></code>	Define a macro.	<code>+define+<marco></code>
<code>+disable_message+<error></code>	Suppress the specified message(s), all error messages.	<code>-error=none</code>
<code>+disable_message+<warning></code>	Suppress the specified message(s), all warning messages.	<code>+warn=none</code>
<code>-disableVYDbg</code>	Disable the knowledge database creation for <code>-v -y</code> files.	

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Vericom to Vlogan Options Mapping Table

<code>-error=no<Error_ID>,...</code>	Report the specified error messages as warning messages. The following error type is supported: MPD: Module <module_name> redefined	
<code>-errormax <number></code>	Specify the maximum number of errors to report. If the errors exceed the number, the parser will skip reporting the remaining errors.	
<code>-errorstop redefined_module</code>	Stop parsing the remaining files when there is a redefined module.	
<code>-extinclude</code>	Compile the included files with the version specified by its extension. When this option is not specified and a source file for one version of Verilog contains the include compiler directive, vericom by default compiles the included file for the same version of Verilog, even if the included file has a different filename extension.	
<code>-extractRTL</code>	Automatically recognizes the RTL storage elements. The extracted storage elements will be saved into the library.	
<code>-f <filename></code>	Load an ASCII file containing design source files and additional simulator options.	
<code>-F <filename></code>	Load an ASCII file containing a specified path to source files and simulator options. Relative path can be used to specify source files.	
<code>-file <filename></code>	Load an ASCII file containing design source files and additional simulator options.	
<code>-ft</code>	Fault tolerance	<code>-Xft=dumpDB</code>
<code>+fixCellHier+<cell_name>+<library_name></code>	Fix the hierarchy resolving reference libraries for specific cell.	

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Vericom to Vlogan Options Mapping Table

-forceIncsaveVY	Force the knowledge database creation for -v -y files in each library module resolution to decrease the compilation memory. This option is valid only when "-incsave" is specified.	
-h	Display this help menu.	-h
-help	Display this help menu.	-help
-ignore <keyword_argument>	Suppress error messages associated with the specified keyword argument. The following keywords are supported: driver_checks Suppress error messages about SystemVerilog driver checking.	
+ignorefileext+<extension_name>	Specify the file extension for ignore.	
-ignorekwd_config	Ignore the keyword of Verilog IEEE 1364-2001, "config".	
-ignore_macro_redef	Suppress warning messages for re-defined macro(s).	
+incdir+<directory_name>	Specify the search path for files used by the `include statement.	+incdir+<directory>
-incdir <directory>	Specify an include directory.	+incdir+<directory>
-incsave	Specify to decrease the compilation memory.	
-L	Specify the library to search for packages.	-liblist
-lib <libName>	Specify the library to save your design to (the default value is "work")	-work
+libext+<extension_name>	Used to specify the file extensions for Verilog library files. See also the -y option.	+libext+<extension>
-libmap <filename>	Specify the library mapping file.	-libmap

<code>+liborder</code>	Search for the module definitions of unresolved module instances with the following order: 1. search the remainder of the library where the unresolved module instances were found. 2. search through the rest of the libraries. 3. search again from the first library.	
<code>+librescan</code>	Search for the module definitions of unresolved module instances by always starting from the first library list specified in the vericom command line.	
<code>+libverbose</code>	Print a message on the screen and in the compiler.log file to indicate the library file where the instance is resolved when a module is instantiated in the source file or library files. The path of the generated lib++ will also be saved into the log file.	
<code>-logdir <directory></code>	Specify the location of the log directory.	
<code>-nclib</code>	Use NC binding scheme. With this option, vericom looks for libMap, workLibrary, viewMap and defaultView in the [NC_Sim] section of the novas.rc for compiling Verilog source files into Novas' KDB.	
<code>+nlog <filename></code>	Append the compile messages to the specified file.	-1
<code>-nlog <filename></code>	Output the compile messages to the specified file.	-1
<code>-ntb_opts ovm[-<version>]</code>	Load the OVM library for compilation. To compile an external OVM library, the VCS_HOME or VCS_OVM_HOME environment variable should be set first.	<code>-ntb_opts ovm[-<version>]</code>

<code>-ntb_opts rvm vmm</code>	Load the VMM library for compilation. To compile an external VMM library, the VCS_HOME environment variable should be set first.	<code>-ntb_opts rvm vmm</code>
<code>-ntb_opts uvm[-<version>]</code>	Load the UVM library for compilation. To compile an external UVM library, the VCS_HOME or VCS_UVM_HOME environment variable should be set first.	<code>-ntb_opts uvm[-<version>]</code>
<code>-nv <filename></code>	Specify a library file to use. Modules in this library file will not be treated as library cells.	<code>-v</code>
<code>-ny <directory></code>	Specify a library directory to use. Modules in this library directory will not be treated as library cells.	<code>-y</code>
<code>-onlylog</code>	Dump the compiler.log only without parsing the design.	
<code>+optconfigfile+<cell_configuration_file></code>	Specify the cell information configuration file that describes which module to potentially replace with which module.	
<code>-ovm[-<version>]</code>	Load the default Verdi OVM library. If -ovm and -ovmhome are specified at the same time, -ovm will be ignored. -ovmhome <path> Specify the OVM installation directory.	
<code>-P <filename></code>	Specify a PLI table file.	<code>-P <filename></code>

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+pkgdir+<pkg_name pkg_full_path>	Load the specified DesignWare VIP package(s) as black box(es) by referring to the headers located in the release package or in an absolute path. For example, > vericom +pkgdir+amba <design> ... -> Compile the amba.f package located in the default product directory. -> Default product directory: "<NOVAS_INST_DIR>/etc/kdb/verilog/dwvip/AMBA_PKG" > vericom +pkgdir+/home/mydwvip/packages/amba <design> ... -> Compile all *.f files in "/home/mydwvip/packages/amba".	
-psl_inline <filename>	Extracts inline-PSL code and stores in specified file.	
-psl_inline_append <filename>	Extracts inline-PSL code and append to a specified file.	
-pslfile <filename>	Treat the specified file as a PSL file. This option will be ignored if the "-assert svvunit" option is not specified.	
-q	Turn on quiet mode.	-q
-quiet	Disable minor (compile and load) messages. Only the total number of errors and warnings are reported.	-q
-rcFile	Specify novas.rc rcFile.	
-realport	Support only the "wreal" keyword in Verilog-AMS. This option also supports the "wrealXState" or "wrealZState" macros for the real x or z states respectively.	-realport
-RegPort	When specified, a reg declaration before the output declaration for the same signal will not generate an error.	

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Vericom to Vlogan Options Mapping Table

<code>-rep[lace]</code>	With this option, vericom removes previously saved Verilog modules in the library before compiling. Without this option, vericom will update existing modules in the library by default.	
<code>+rmkeyword+<keyword(s)></code>	Downgrade the specified keyword(s) to identifier(s). One or more keywords may be specified.	
<code>-rmkeyword <keyword></code>	Downgrade the specified keyword to an identifier.	
<code>-saveOpts</code>	Save the compilation options.	
<code>-silent</code>	Prevent bells (ctrl-G) from being issued.	
<code>+spiceext+<extension_name></code>	Specify the file extension for HSpice files.	
<code>-sort_net_instance</code>	Sort the signals and instances alphabetically.	
<code>-stdout</code>	Print compiler messages to stdout.	Default
<code>-ssv</code>	Library modules in the library file (-v) will not be tagged as library cells.	<code>-Xvd_opts=-ssv</code>
<code>-ssy</code>	Library modules in the library directory (-y) will not be tagged as library cells.	<code>-Xvd_opts=-ssy</code>
<code>-ssz</code>	Ignore the `celldefine compiler directives.	<code>-Xvd_opts=-ssz</code>
<code>-sv</code>	Support SystemVerilog IEEE 1800-2005 standard.	<code>-sverilog</code>
<code>-sverilog</code>	Support SystemVerilog IEEE 1800-2005 standard.	<code>-sverilog</code>
<code>-sv_pragma</code>	Compile the SystemVerilog assertions code that follows the sv_pragma keyword in a comment.	<code>-sv_pragma</code>

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<code>-syntaxerrormax <number></code>	Specify the maximum number of syntax errors to stop parsing. If the syntax errors exceed the number, the parser will stop parsing the remaining files.	
<code>+systemverilogext+<extension_name></code>	Specify the file extension for SystemVerilog files.	<code>+systemverilogext</code>
<code>-timescale=<time_scale></code>	Set the default timescale for the modules without timescale definition.	<code>-timescale = time_unit/time_precision</code>
<code>-u[ppercase]</code>	Change all identifiers to uppercase.	
<code>-useius=adv</code>	Use IUS style parsing/elaboration.	<code>-kdb=useius</code>
<code>-usemti=adv</code>	Use MTI style parsing/elaboration.	<code>-kdb=usemti</code>
<code>-usevcs=adv</code>	Use VCS style parsing/elaboration.	<code>-usevcs=adv</code> is input option
<code>-uvm[-<version>]</code>	Load the default Verdi UVM library. If -uvm and -uvmhome are specified at the same time, -uvm will be ignored.	
<code>-uvmhome <path></code>	Specify the UVM installation directory.	
<code>-v <filename></code>	Modules in the specified file will be treated as library cells.	<code>-v</code>
<code>-v95</code>	Support Verilog IEEE 1364-1995 standard.	<code>-v95</code>
<code>+v2k</code>	Support Verilog IEEE 1364-2001 standard.	<code>+v2k</code>
<code>-vc</code>	Support DirectC syntax.	<code>+vc</code>
<code>+verilog2001ext+<extension_name></code>	Specify the file extension for Verilog 2001 files.	<code>+verilog2001ext+</code>
<code>+v95ext+<extension_name></code>	Specify the file extension for Verilog 1995 files.	<code>+verilog1995ext+</code>
<code>+verilog1995ext+<extension_name></code>	Specify the file extension for Verilog 1995 files.	<code>+verilog1995ext+</code>

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<code>-view <viewName></code>	Specify NC viewName. This option is valid only when "-nclib" is specified. This provides the same function as ncvlog -view view_name to use the specified view name for the design units of the input Verilog source files. For example: vericom -nclib -view novas ../src/novas.v All design units in ../src/novas.v have a view name novas. Note: Using the command line option, -view, overrides defaultView and viewMap variables in novas.rc.	
<code>-wcFile</code>	Support the wildcard file list in a run.f file. Note that the /*...*/ comment syntax is not supported in the run.f file.	
<code>-work <libName></code>	Specify NC libName. This option is valid only when "-nclib" is specified. This provides the same function as ncvlog -work library_name to use the specified library name for the design units of the input Verilog source files. For example: vericom -nclib -work novas -view novas -f ../src/novas.f All design units in ../src/novas.f are compiled into the library novas and have a view name novas. Note: Using the command line option, -work, overrides workLibrary and libMap variables in novas.rc.	<code>-work</code>
<code>-y <directory_name></code>	Modules in the specified directory will be treated as library cells.	<code>-y</code>

Note:

If there exists vericom-only parameter, show the warning message:

[Warning] Option <op> is ignored because it is not supported in vericom -usevcs=adv.

[Warning] Options <op>, <op>, <op>, ... are ignored because they are not supported in vericom -usevcs=adv.

Vhdlcom to Vhdlan Options Mapping Table

All vhdlcom options mapped for vhdlan would be allowed as following:

VHDLCOM OPTIONS	DESCRIPTION	VHDLAN OPTIONS
-XOptions="..."	For vericom/vhdlcom/elabcom in advanced mode flow to pass the related options without mapped.	Related vlogan/vhdlan/vcs options
-87	Supports the VHDL IEEE 1076-1987 standard.	-vhdl87
-comment_transoff_regions [+<synopsys>]	Skips the source code between <code>translate_off</code> and <code>translate_on</code> pragmas. For examples, see the <code>-comment_transoff_regions</code> option in the verdi utility description.	-skip_translate_body [=synopsys]
-f -file -fv filename	Specifies the run file. If there are several VHDL files to compile, pack them sequentially into a file, then pass this file to <i>vhdlcom</i> . Example: If three source files called <i>v1.vhd</i> , <i>v2.vhd</i> , and <i>v3.vhd</i> exist, create a file called <i>run.f</i> (or another desired name) and create its contents as follows (note that each file name must occupy its own line): <i>v1.vhd v2.vhd v3.vhd</i>	-f
-h -help	Prints the help message.	-h -help
-lib libName	Specifies the library to save the design. The default library is <i>work</i> .	-work libName
+libverbose	Prints a message on the screen and in the <i>compiler.log</i> file to indicate the path of the generated <i>lib++</i> file.	-verb[ose] [off]
-nlog filename	Outputs the compile messages to the specified file.	-l
+nlog filename	Appends the compile messages to the specified file.	-l
-psl_inline <filename>	Extract inline-PSL code and store it in the specified file.	-pslfile filename

-q	Turns on the quiet mode to suppress the console information of copyright, version, locations of the log directory, resource file, and GUI configuration file.	-nc (Suppress copyright banner)
-quiet	Disables minor (compile and load) messages. Only the total number of errors and warnings are reported.	-q
-rcFile <i>rcFilename</i>	Specifies the resource file. Its default value is <i>novas.rc</i> .	use SYNOPSISYS_SIM_SETUP
-saveOpts	Save the compilation options to the Verdi Knowledge Database library.	
-silent	Prevents bells (Ctrl-G) from being issued.	
-skip <i>eapbc</i>	Directs the compiler to skip the specified type of design units (that is, <i>e</i> : entity, <i>a</i> : architecture, <i>p</i> : package, <i>b</i> : package body, <i>c</i> : configuration). For example, - <i>skip e</i> does not compile entities and - <i>skip ea</i> does not compile entities and architectures.	
-smartorder	Compiles in order independent mode.	-smart_order
-smartscrip <i>filename</i>	Generates a compilation script file with the specified name.	-smartscrip
-vhdl08	Supports the VHDL IEEE 1076-2008 standard.	-vhdl08
-work libname -w libname	Specifies the library to save the design. The default library is <i>work</i> .	-work <i>libname</i>
-useius -cadence	set simulator type	-useius=adv
-usemti -Mentor	set simulator type	-usemti=adv
-usevcs -synopsys	set simulator type	-usevcs=adv

Elabcom to VCS Elaboration Options Mapping Table

ELABCOM OPTIONS	DESCRIPTION	VHDLAN OPTIONS
-XOptions="..."	For vericom/vhdlcom/elabcom in advanced mode flow to pass the related options without mapped.	Related vlogan/vhdlan/vcs options
-gv -gvalue generic=<value>	Overrides the value defined in the source code with the value specified in the option.	-gv <generic_name>=<value>
-L/-Lf	Specify one or more Verdi libraries compiled with vericom or vhdlcom that do not contain the top view. -Lf has higher priority than -L.	-liblist
-elab <path>	Specify the path of the generated Verdi elabDB.	- Xverdi_elab_opts=-elab, <path>
-impConf <confFile>	Specify a file to configure how undefined modules, virtual tops and partial loads are handled during design import. This option will not load a VHDL design import. This option will not load a VHDL configuration file.	-Xpartialdesign
-lib <libName> -top<topName>	Specify library name.	<libName>.<topName>
-liblist	searches the specified libraries compiled by VCS. The plus sign (+) is used to include multiple libraries in the library list.	-liblist
-liblist_work	Starts searching from the parent library first. The -liblist_work option has higher search priority than the -liblist option.	-liblist_work
-liblist_nocellldiff	in library searching for a master name, use the first found target and don't care it's library cell or design cell	-liblist_nocellldiff
-novasLibPaths <filename(s)> <libname(s)>	Specifies one or more symbol library paths or files containing the paths to all symbol libraries.	- Xverdi_elab_opts=-novas LibPaths,<files>

<code>-novasLibs <filename(s)> <libname(s)></code>	Specifies one or more symbol library paths or files containing the names to all symbol libraries.	<code>-Xverdi_elab_opts=-novasLibs,<files></code>
<code>-ovm[<-version>]</code>	Loads the default Verdi OVM library. NOTE: If -ovm and -ovmhome are specified at the same time, -ovm is ignored.	<code>-ntb_opts ovm \$VCS_OVM_HOME/src/verdi/dpi/ovm_verdi_dpi.cpp ovm_custom_install_verdi_recording</code>
<code>-ovmhome <path></code>	Specifies the OVM installation directory	<code>-ntb_opts ovm[<-version>] \$VCS_OVM_HOME/src/verdi/dpi/ovm_verdi_dpi.cpp ovm_custom_install_verdi_recording</code>
<code>+optconfigfile+<cell_configuration_file></code>	Specify the cell information configuration file that describes which module to potentially replace with which module.	<code>+optconfigfile+<filename></code>
<code>-parameters <filename></code>	Specifies the file containing a list of parameters to be changed to values. The file format is assign <value> <parameter_path>	<code>-parameters <filename></code>
<code>-pvalue+<parameter_hierarchical_name>=<value></code>	Change the specified parameter to the specified value.	<code>-pvalue+<parameter_hierarchical_name>=<value></code>
<code>-ssv</code>	Do not automatically tag library modules in the library directory (-v) as library cells.	<code>-Xvd_opts=-ssv</code>
<code>-ssy</code>	Do not automatically tag library modules in the library directory (-y) as library cells.	<code>-Xvd_opts=-ssy</code>
<code>-ssz</code>	Ignore `celldefine compiler directive.	<code>-Xvd_opts=-ssz</code>
<code>-uvm[<-version>]</code>	Loads the default Verdi UVM library. NOTE: If -uvm and -uvmhome are specified at the same time, -uvm is ignored.	<code>-ntb_opts uvm<-version> -native_uvm_dumping uvm[<-version>]</code>
<code>-uvmhome <path> -uvm[<-version>]</code>	Specifies the UVM installation directory with native dumping.	<code>-ntb_opts uvm<-version> -native_uvm_dumping <-version></code>

<code>-uvmhome <path></code>	Specifies the UVM installation directory.	<code><path>/src/dpi/uvm_dpi.cc -CFLAGS -DVCS</code>
<code>+disable_message+error</code>	Suppresses all error messages.	<code>-error=none</code>
<code>+disable_message+warning</code>	Suppresses all warning messages.	<code>+warn=none</code>

Libraries Mapping

Vericom/Vhdlcom will transform the libraries mapping from vericom/vhdlcom(novas.rc) to vlogan/vhdlan (synopsys_sim.setup).

```
novas.rc
-----
[Library]
work = ./work
A = ./a

to

synopsys_sim.setup
-----
A: ./a
```